Credit Card Customer Behavior

Hypothesis Testing + Clustering Extension

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Meet the Team

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Project Overview

Research Question

Do distinct patterns of credit card usage carry statistically measurable differences, and can we identify natural customer profiles through clustering?

	Part 1	**Part 2**
Approach	Hypothesis Testing	K-Means Clustering
Tools	`t.test()`, `cor.test()`	`kmeans()`, `dplyr`
Goal	Confirm or debunk claims	Discover customer groups
Dataset	CC GENERAL — 8,950 customers, 18 variables	

Significance level for all tests: $\alpha = 0.05$

Data Wrangling with dplyr

```
cc_clean <- cc %>%
  filter(!is.na(CREDIT_LIMIT), !is.na(MINIMUM_PAYMENTS)) %>%
  mutate(
    CA_Group    = if_else(CASH_ADVANCE_FREQUENCY >= 0.25,
                          "Heavy CA", "Light CA"),
    LOG_BALANCE = log1p(BALANCE)
  )
```

```
Rows after cleaning: 8636
```

```
Heavy CA Light CA
    2204     6432
```

Part 1: Hypothesis Testing — Overview

H1 – Heavy Cash-Advance Users Carry Higher Balances

Test: One-tailed Welch's two-sample t-test **Expected: Confirmed**

H2 – Balance Strongly Predicts Credit Limit (Debunk)

Test: Pearson correlation `cor.test()` **Expected: Debunked**

	H1	H2
Null Hypothesis	$\mu_{\text{Heavy}} = \mu_{\text{Light}}$	$\rho = 0$
Alternative	$\mu_{\text{Heavy}} > \mu_{\text{Light}}$	$\rho \neq 0$ (strong)
Key metric	t-statistic, p-value	r, r^2

H1: Formal Statement

$H_0 : \mu_{\text{Heavy CA}} = \mu_{\text{Light CA}}$ ($\mu_{\text{Heavy CA}}$ is the average balance of Heavy CA users, and it is equal to $\mu_{\text{Light CA}}$ which is the average balance of Light CA users)

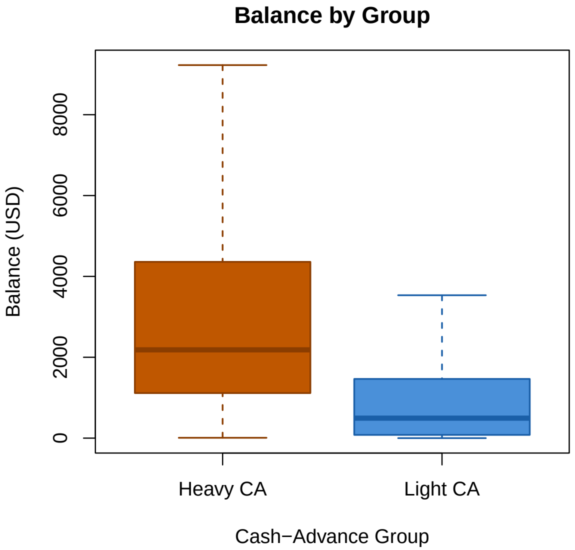
$H_1 : \mu_{\text{Heavy CA}} > \mu_{\text{Light CA}}$ ($\mu_{\text{Heavy CA}}$ is the average balance of Heavy CA users, and it is greater than $\mu_{\text{Light CA}}$ which is the average balance of Light CA users)

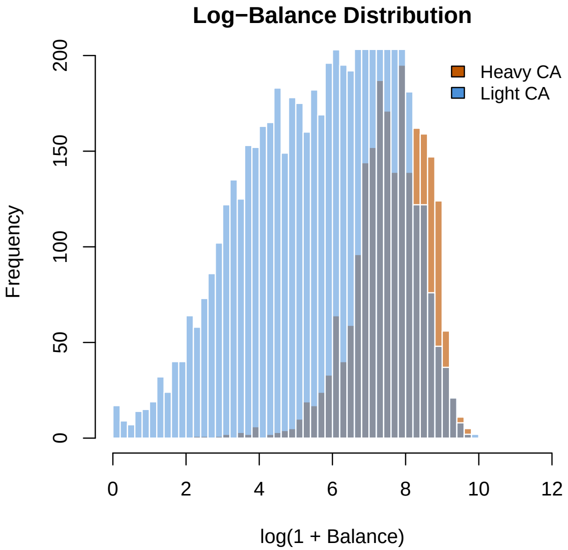
Rationale

The correlation matrix shows $r(\texttt{BALANCE}, \texttt{CASH\_ADVANCE\_FREQUENCY}) \approx 0.449$. Customers who rely on cash advances tend to accumulate higher account balances (`BALANCE` = balance amount left in account to make purchases) — because cash advances incur high fees and interest immediately upon withdrawal.

Test chosen: Welch's two-sample t-test (`var.equal = FALSE`) — safe default when group variances differ.

H1: Exploratory Plots





H1: Welch's *t*-Test

```r
t_result_h1 <- t.test(
  x           = heavy,
  y           = light,
  alternative = "greater",   # one-tailed: Heavy > Light
  var.equal   = FALSE        # Welch's t-test
)
```

```
t  = 32.319

df = 2889.6

p  = 3.43e-196

95% CI (lower bound): $1775
```

H1: Decision & Interpretation

```
t-statistic : 32.319
p-value     : 3.43e-196
95% CI lower bound: $1775.11
Decision    : REJECT H0
```

Interpretation

The large positive t and the near-zero p-value provide strong statistical evidence: heavy cash-advance users carry **significantly higher** balances on average. CA frequency is a **measurable financial risk signal**.

H2 (Debunk): The Intuitive Claim

Common Intuition

"Customers with higher balances must have higher credit limits — the bank gave them more credit, so they use more of it."

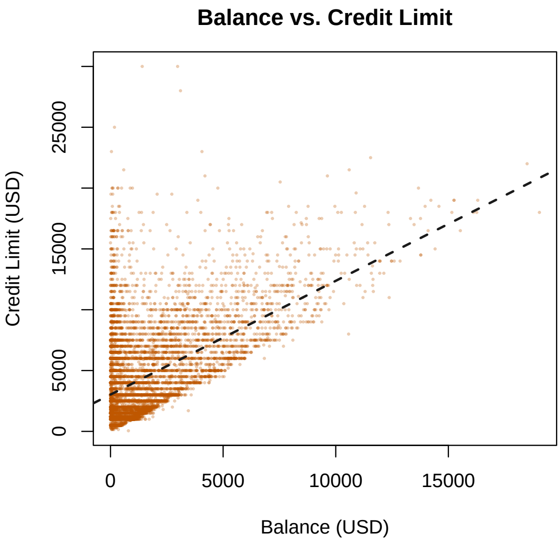
$\boldsymbol{H_0 : \rho = 0}$ (ρ is the correlation between BALANCE and CREDIT_LIMIT, and it is equal to zero, meaning there is no relationship)

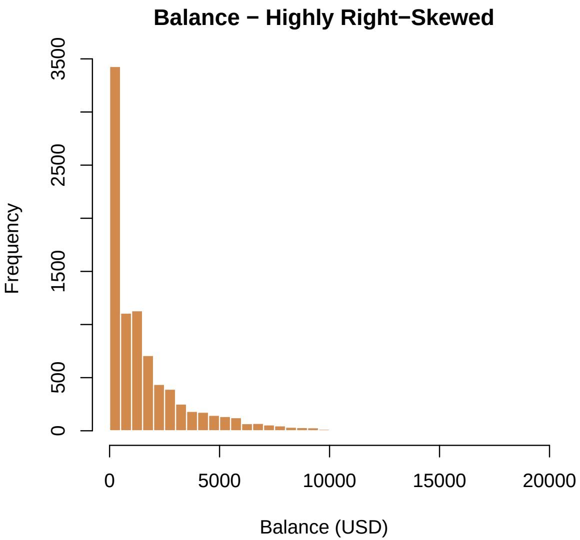
$\boldsymbol{H_1 : \rho \neq 0}$ (ρ is the correlation between BALANCE and CREDIT_LIMIT, and it is not equal to zero, meaning there is a strong positive relationship)

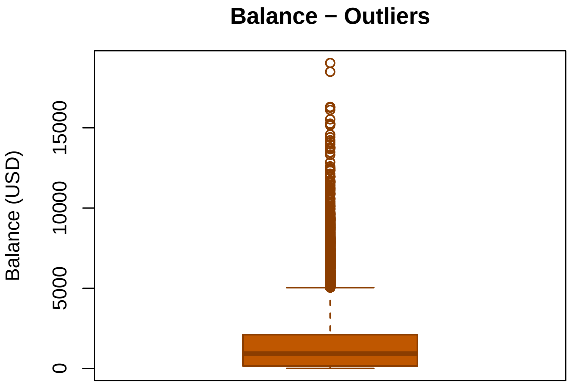
Debunk strategy — three prongs:

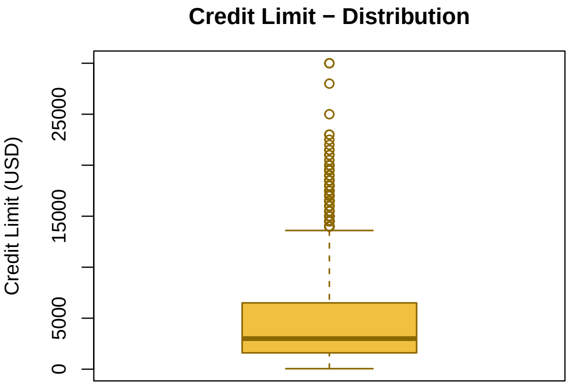
1. **Moderate** r from `cor.test()` — r^2 explains only $\approx$ 29% of variance
2. **Severe outliers** in `boxplot()` that distort the visual pattern
3. **Right-skewed distribution** in `hist()` — normality assumption violated

H2: Scatterplot & Distribution









`Balance outliers (IQR rule): 666 customers`

H2: Pearson Correlation Test & Debunk

```r
cor_h2 <- cor.test(cc_clean$BALANCE, cc_clean$CREDIT_LIMIT,
                   method = "pearson")
```

```
r (Pearson)             : 0.5355
r^2 (variance expl.): 28.68%
p-value                 : 0.00e+00
|r| < 0.35?             : NO  -> DEBUNKED
```

Key Lesson: Significance $\neq$ Practical Strength

With $n = 8{,}950$, even a moderate correlation gives a near-zero p-value. The actual $r \approx 0.54$ seems moderate, but $r^2 \approx 0.29$ means BALANCE explains only $\approx$ 29% of CREDIT_LIMIT variance. Banks set limits *prospectively* (based on income and credit history), not in response to current balance — the strong-prediction claim is **false**.

Key Correlations Found

Three most important relationships

- **PURCHASES and PAYMENTS** $r \approx 0.60$ (strong positive) — customers who spend more also tend to pay more.
- **CASH_ADVANCE and BALANCE** $r \approx 0.50$ (moderate positive) — heavier cash-advance use is associated with a higher account balance.
- **PRCFULLPAYMENT and BALANCE** $r \approx -0.32$ (negative) — customers who pay in full carry much lower account balances.

The matrix also revealed high redundancy between some pairs (e.g., PURCHASES and ONEOFF_PURCHASES: $r \approx 0.92$), which guided the selection of non-overlapping variables for clustering.

Correlation Matrix

Correlation Matrix –– Clustering Variables

	BAL	PURCH	CA	CL	PAY	PF	CAF	FULL
BAL	1.00	0.18	0.50	0.54	0.32	−0.09	0.45	−0.33
PURCH	0.18	1.00	−0.05	0.36	0.61	0.39	−0.12	0.18
CA	0.50	−0.05	1.00	0.30	0.46	−0.22	0.63	−0.16
CL	0.54	0.36	0.30	1.00	0.43	0.12	0.13	0.05
PAY	0.32	0.61	0.46	0.43	1.00	0.10	0.18	0.10
PF	−0.09	0.39	−0.22	0.12	0.10	1.00	−0.32	0.31
CAF	0.45	−0.12	0.63	0.13	0.18	−0.32	1.00	−0.26
FULL	−0.33	0.18	−0.16	0.05	0.10	0.31	−0.26	1.00

Complementary Question:

The hypothesis tests confirmed specific claims about individual variables.

Now we ask: *Can we find natural customer groups in the data?*

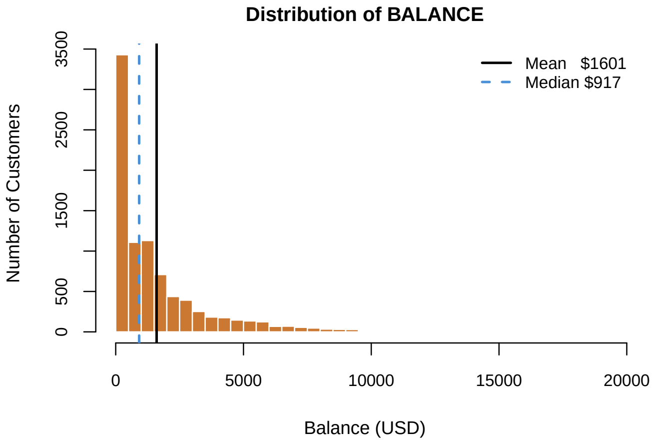
Method: K-Means Clustering — an exploratory, unsupervised technique

Input: 8 financial variables, standardised

Result: 5 interpretable customer profiles

EDA: Key Findings

- **8,950 customers**, 18 variables
- All major financial variables are **strongly right-skewed**:
  - PURCHASES: skew = 8.14
  - CASH_ADVANCE: skew = 5.17
  - BALANCE: skew = 2.39
- Mean $\gg$ Median for most variables — a small group of high-activity customers pulls the average up
- Extreme values **kept** — they represent real, meaningful customer behavior



Why K-Means Clustering?

The dataset has no predefined customer categories

The EDA showed that customers behave very differently from one another. Clustering groups observations that share similar characteristics, without requiring a predefined label.

- **8 variables** selected: BALANCE, PURCHASES, CASH_ADVANCE, CREDIT_LIMIT, PAYMENTS, PURCHASES_FREQUENCY, CASH_ADVANCE_FREQUENCY, PRCFULLPAYMENT
- Variables were **standardised** so that dollar-valued variables do not dominate frequency variables (0 to 1 scale)
- **K = 5** chosen by comparing several values — five groups provide a useful balance between separation and interpretability
- K-Means is used here as an **exploratory complement** to the hypothesis-testing analysis above

K-Means Result: 5 Customer Groups

Cluster Name	% Customers	Key Characteristics
Low-Activity Customers	38.7%	Very low purchases, low purchase frequency
Regular Purchasers	32.7%	Frequent purchases, very low cash-advance use
Full-Payment Customers	14.8%	Pays full balance consistently, lowest debt carried
High Cash-Advance Users	12.4%	Very high balance driven by cash advances
High-Value Customers	1.3%	Extremely high spending and credit limit ($n = 112$)

Conclusions

Part 1: Hypothesis Testing

- **H1 Confirmed** — Heavy cash-advance users carry significantly higher balances ($p \ll 0.05$). CASH_ADVANCE_FREQUENCY is a reliable credit-risk signal.
- **H2 Debunked** — Despite $p < 0.05$, BALANCE explains only $\approx 29\%$ of CREDIT_LIMIT variance ($r \approx 0.54$, $r^2 \approx 0.29$). Statistical significance does not imply practical strength.

Part 2: Clustering Extension

K-Means identified **five interpretable customer profiles** that summarise different financial behaviors across the 8,950 customers. These are **exploratory profiles, not official categories** — they complement and contextualise the hypothesis-testing findings above.

References

- **Dataset:** CC GENERAL — Kaggle Credit Card Customer Segmentation (8,950 obs.)
- **R Packages:** `dplyr` (wrangling) · Base R (`stats`, `graphics`)
- **Statistical Functions:** `t.test()` · `cor.test()` · `kmeans()`
- **Visualization:** `boxplot()` · `hist()` · `plot()` · `barplot()` · `image()`
- **Concepts Applied:**
  - Welch's Two-Sample t-Test
  - Pearson Correlation — Effect Size vs. Significance
  - K-Means Clustering (K $=$ 5, standardised variables)
  - Outlier Detection via IQR Rule

Project Access

Scan for Full Project Materials

Full report, presentation source, dataset, and study guide:

`github.com/Sbr0401/FinalBigDataUTEXAS`